BITEBOUND

Simon Völkl, Johannes Schieder
Mobile and Ubiquitous Computing | OTH Amberg-Weiden

Repository: github.com/s-voelkl/BiteBound

Motivation

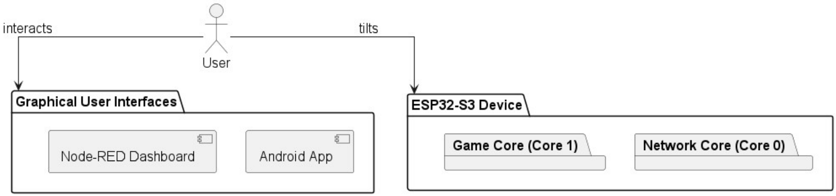
Ziel: Hoch performante IoT **Architektur** für ESP32-Spiel

Risiko hoher **Latenz**

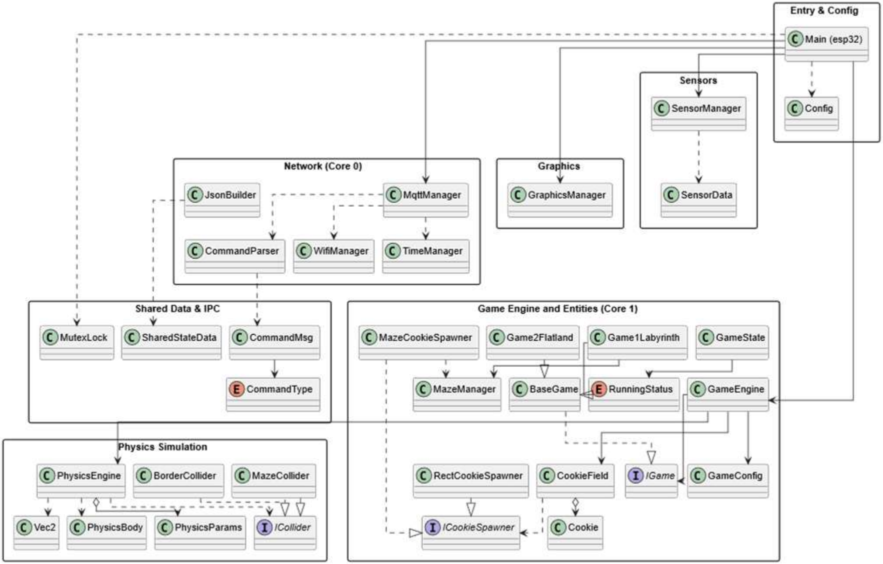
- Game & Physik Engine
- MQTT-Kommunikation
- Display

Prozessverteilung auf zwei Cores (dual-core)

System Architektur

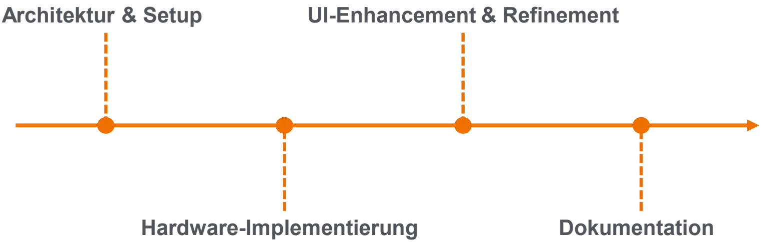


Hardware Architektur

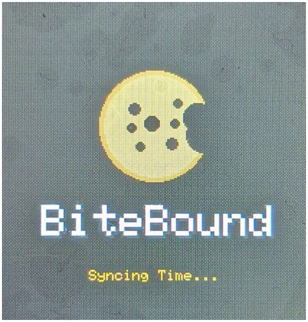


Hardware Architektur

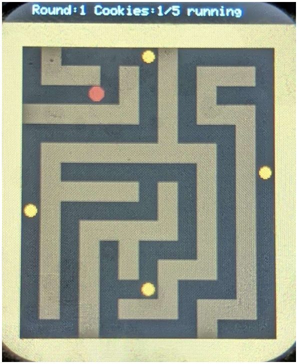
4 Meilensteine



ESP32



Loading Screen



Loading Screen

Android

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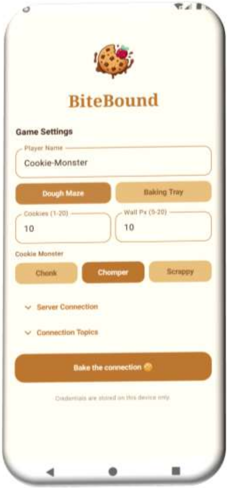
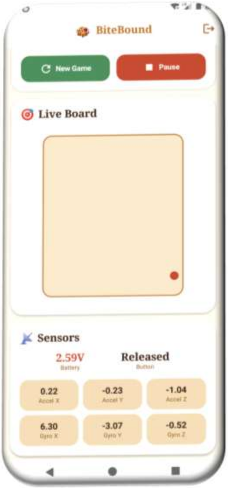
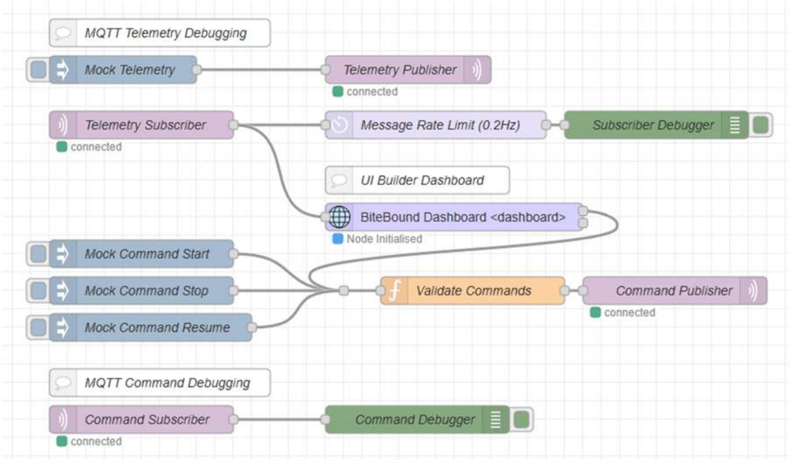




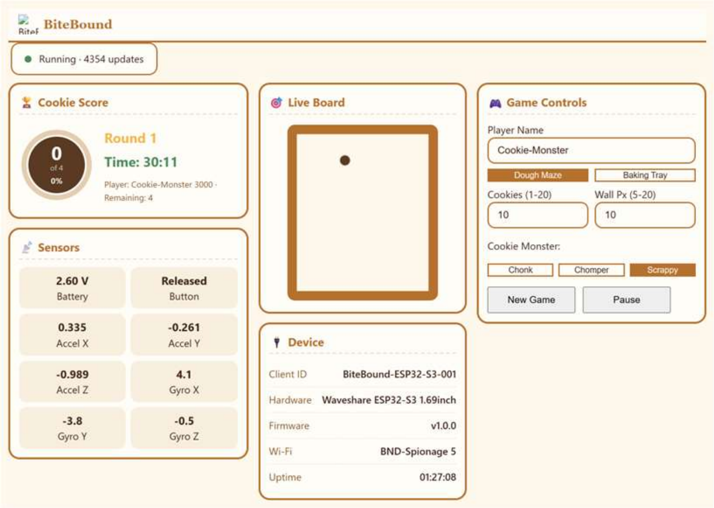
Node-RED

Workflow



Node-RED

UI Builder Dashboard



Code Qualität

Hardware & Frontend

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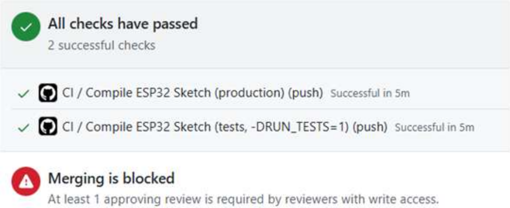
Tests:

- Hardware (Arduino AUnit): **61 Unit Tests**
- Android Tests: **82%** Line-Coverage auf Data-Layer
- Mocks für MQTT-Nachrichten in Node-RED
- Kritisches Feedback durch Peer-Testing

GitHub Actions:

- CI/CD Pipelines
- Pull Request Requirement



Future Work

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- Speichern der Daten (z.B. Leaderboards)
- Sound und Haptik
- Button-Interaktion mit Spezial-Aktionen

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